

A Symbolic Substrate for Computational Materials Science: Principles and First-Demonstration Plan

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Abstract

This document records the design of a typed substrate for computational materials science workflows, organized around a small set of principles. Workflows are directed acyclic graphs of *abstract physics states* connected by *symbolic operations*, and concrete numerical results are *materializations*—typed projections of abstract states into a discretization scheme. The substrate’s abstract root is the *Born–Oppenheimer potential* of a material, expanded as a Taylor series in atomic displacements; force constants of every order are derived states obtained by extracting derivatives of this root and truncating the expansion at a chosen order. Every state carries a provenance chain that records the operations along its path; every materialization carries a discretization scheme. Approximation and discretization errors live in disjoint type universes and compose along graph edges; the machinery for tracking error formulas symbolically and composing them across the graph is designed but its full implementation is deferred to Phase 2. As a first demonstration, three established lattice thermal-transport codes (kaldo, phono3py, ShengBTE) are reframed as distinct materializations of a single abstract DAG, with cross-code reconciliation at canonical intermediate states as the Phase 1 deliverable. Phase 1 stubs the upstream operations that produce force constants from the potential—declaring the relevant abstract types and provenance labels but not modeling them in detail—so that Phase 2 can elaborate the upstream without restructuring the substrate. The document is intended as the working architectural reference for the `openmaterials-ai` project; it covers each architectural decision, the alternatives considered, and what we deferred.

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1 Principles

The substrate is built on a small set of commitments. They are stated here without their alternatives or justifications; supporting definitions follow in later sections.

1. **Two worlds, one bridge.** The substrate keeps the abstract physics world (typed states, symbolic operations, approximation errors) strictly separate from the numeric world (discretized arrays, computed values, discretization errors). They are connected by exactly one bridge: the materialization functor.
2. **Abstract states are unit-free.** Dimensionful quantities (frequency, energy, length, time) exist at the abstract layer, but specific unit choices (THz vs eV vs rad/s, Å vs Bohr) do not. Unit choice belongs to the materialization and is declared by the adapter that produced it; the substrate normalizes to a canonical unit before any cross-materialization comparison.
3. **The atom is a symbolic operation.** The atomic unit of meaning is a typed transformation between abstract states, not an instruction in an input file or an algorithm. Every operation carries a *symbolic formula*—a sympy expression (closed-form output) or sympy.Eq (implicit equation defining the output)—stating what it computes. The formula is the substrate’s claim, machine-readable and Lean-projectable, against which adapter conformance can be checked.
4. **States are typed witnesses; equivalently, observables are nodes.** A state is the typed claim that an abstract physics object exists with a particular history—not the data the object would have when computed. In practice each state corresponds to a single observable, so the substrate uses “state”, “observable”, and “node” interchangeably. Each observable declares an index signature (e.g. $\omega(\mathbf{q}, \nu)$ has indices $(\mathbf{q}, \nu)$; $\kappa^{\alpha\beta}$ has (α, β)). Numerical content lives in the materialization, separate from the witness.
5. **Workflows are directed acyclic graphs.** A workflow is a graph of states connected by operations, not a linear sequence. Cross-code comparison reduces to graph alignment over the same abstract template.

6. **Identity is type plus provenance.** An abstract state is identified by its physics type and the ordered chain of operations that produced it. Two states with the same type but different histories are different states.
7. **Operation identity is parameterized.** An operation is identified by its name together with the parameters that affect physics. Different physics-affecting parameter values produce different operations and therefore different output states. Function-family parameterizations (e.g., Gaussian broadening by standard deviation vs. by half-width) are canonicalized at the abstract layer to a single choice; adapters translate to their internal convention.
8. **Discretization belongs to materialization.** Mesh densities, supercell sizes, displacement steps, and broadenings live on the materialization, not on the abstract state. Convergence is a relationship between an abstract state and a family of its materializations.
9. **Cross-code agreement is at the observable layer.** Two materializations of the same abstract state produced by different adapters need not agree numerically at intermediate decompositions (per-mode linewidths, basis-dependent intermediate sums); they must agree at extracted observables. Each abstract observable carries a comparison protocol declaring its expected tolerance across adapters. Contracted scalars and tensors (e.g., $\kappa^{\alpha\beta}$, $\Sigma\Gamma_{q\nu}$) carry tight protocols; per-mode arrays whose values depend on summation order or symmetry-reduction choices carry loose or not-comparable protocols.
10. **Errors are designed-for.** Approximation errors live on operations; discretization errors live on materializations. The type system reserves slots for both; symbolic composition of error formulas is a Phase 2 capability.
11. **The abstract root is the potential.** Every materialization in the substrate ultimately traces back to the Born–Oppenheimer potential of the material. Force constants of every order are derived states obtained by extracting derivatives of the potential and truncating the Taylor expansion. The upstream is stubbed in Phase 1.
12. **Sources are produced by nullary operations.** Source states (the potential, but also temperature, lattice constant, etc.) are the outputs of nullary `provide_*` operations rather than nodes with empty provenance. This unifies the DAG: every node is the output of some operation, and the substrate has no special “input” category. The materialization of a source state carries the externally supplied value (e.g., $T = 300\text{ K}$); the operation is just a structural tag.
13. **Lean-compatible by structure.** The substrate is implemented in Python but designed so that its types transliterate cleanly into Lean. Three disciplines protect this option: closed unions for physics types, no Python-encoded physics invariants, operation parameters always recorded in output state metadata.

2 Background and Motivation

Modern computational materials science has many codes, many workflows, and many opinions about how to compose them. Despite a generation of effort on workflow managers (AiiDA, ATOMATE, FIREWORKS, ATOMATE2, AFLOW), one quality is still missing: a workflow should expose, as typed and composable data, *the physics it computes*.

Today the standard unit of shared meaning between codes is the *input file*: an ordered sequence of instructions that, together with a particular code’s internal logic, produces an output. This unit is opaque to two questions:

- **Given a result, what physical assumptions and approximations does it embed?** An input file declares parameters but not their meaning. A k-mesh is just a triple of integers; the relationship between that mesh and the convergence of the integrated observable is not part of the file.
- **Given two results from two codes, where exactly do they diverge?** Identical force constants run through kaldo and through ShengBTE typically yield slightly different thermal conductivities. The difference is not localized to a particular operation, even though—in principle—it is the result of distinct computational choices at specific steps.

A previous attempt of ours, **MaterialsCodeGraph** (MCG), addressed parts of this gap by treating each instruction in an input file as the atomic unit of a knowledge graph and tracking inputs, parameters, and outputs at that level. MCG’s tool-encapsulation principle (all tool knowledge in YAML/Markdown configs, all code generic over tool) was the right design at that level of abstraction. But the level of abstraction itself is too low: an instruction in an input file is a syntactic unit, not a semantic one. It tells you what a code is told to do, not what *physics* the code thereby computes. Two codes that compute the phonon dispersion via diagonalization of the dynamical matrix express that operation as different instructions; yet they perform the same physics. A graph indexed by instructions sees them as two unrelated tasks.

The substrate proposed here *starts from the physics* and treats input files as derived artifacts. The atomic unit is no longer an instruction; it is a *symbolic operation* between typed physics states. The states are the load-bearing entities: a state is what changes when an operation is applied, and a workflow is the graph that records those state transitions. Concrete numerical computation is then a separate, typed projection of the abstract workflow into a discretization scheme.

This is a deliberate departure from the input-file paradigm. We argue below that it pays for the extra abstraction with three concrete returns: cross-code reconciliation at canonical intermediate states, derived (rather than empirical) error bounds on computed observables, and a clean substrate for future formal verification.

3 Architectural Principle: Two Worlds, One Bridge

The single most important design decision is a strict separation between two type universes:

- **The abstract world** holds typed physics states (Hamiltonians, dispersion relations, scattering rates, distribution functions, partition functions) and symbolic operations between them. Errors here are *approximation errors*—the $O(\cdot)$ terms introduced by truncating perturbation series, linearizing equations, or imposing physical assumptions such as the Born–Oppenheimer or harmonic approximation. The abstract world contains no numerical content. Two abstract states that differ only in their numerical estimates are the same abstract state; two abstract states that differ in the chain of approximations leading to them are different abstract states.
- **The numeric world** holds discretized realizations of abstract states: arrays sampled on finite grids, mode-resolved quantities at finite k-meshes, distributions evaluated at finite temperature. Errors here are *discretization errors*—the $O((1/N)^k)$ or $O(h^k)$ terms introduced by sampling a continuum quantity on a finite scheme. The numeric world contains no symbolic content; it does not know what theory the numbers are estimates of.

The two worlds are connected by exactly one bridge, the *materialization functor*, which takes (i) an abstract state and (ii) a discretization scheme and produces a numeric materialization. Approximation errors compose in the abstract world along provenance chains; discretization

errors compose in the numeric world along execution graphs; the two combine only at materialization time, where the total error of a numerical result decomposes into its abstract and numeric contributions.

This separation is load-bearing. It is also somewhat unusual. Most computational physics codes mix the two worlds: a “Hamiltonian” object in such a code is sometimes a typed mathematical operator and sometimes a 4-D `numpy` array. The substrate refuses to do this. If you are holding a Hamiltonian in the substrate, you know which world you are in and you cannot accidentally mix the two. This refusal is what makes errors composable, theorems statable, and cross-code comparisons well-defined.

4 Formal Definitions

We collect the formal definitions implied by the decisions above.

4.1 Abstract state

An abstract state s is unit-free: its physics type carries the dimension of any quantity it represents (a frequency, an energy density, a tensor of given rank), but no unit choice. Numerical content and units appear only at materialization time.

An abstract state s is a tuple $(\tau, \pi, \mathcal{I})$ where:

- τ is a *physics type* drawn from a finite registry: derived types include `ForceConstants[order= $n$ ]`, `DynamicalMatrix`, `Frequency`, `Eigenvectors`, `GroupVelocity`, `HeatCapacity`, `Linewidth`, `MeanFreeDisplacement`, `ThermalConductivity`; source types include `Potential` (the abstract root, stubbed in Phase 1), `Temperature`, `LatticeConstant`, `Pressure`, and measured-observable types;
- π is the *provenance*: an ordered list $[\mathcal{O}_1, \mathcal{O}_2, \dots, \mathcal{O}_n]$ of symbolic operations whose composition produced s . Provenance is never empty—source states have a single nullary operation (`provide_potential`, `provide_temperature`, ...);
- $\mathcal{I}$ is the *index signature*: the tuple of named indices the observable carries (e.g., ω has $\mathcal{I} = (\mathbf{q}, \nu)$; v has $(\alpha, \mathbf{q}, \nu)$; κ has (α, β)). Empty for scalar / opaque observables.

Two abstract states are equal iff $\tau_1 = \tau_2$ and $\pi_1 = \pi_2$. The composed approximation error on s (a Phase 2 capability) would be

$$\epsilon_{\text{approx}}(s) = \text{compose}(\epsilon_{\mathcal{O}_1}, \epsilon_{\mathcal{O}_2}, \dots, \epsilon_{\mathcal{O}_n}). \quad (1)$$

Source versus derived states. Abstract states fall into two kinds, both produced by some operation:

- *Source states* are the outputs of *nullary* operations (`provide_potential`, `provide_temperature`, ...): operations with no abstract-state inputs. They are empirical inputs (lattice constant, temperature, pressure, applied field), the abstract `Potential` itself (Phase 1 stubs the upstream of the potential), or measured observables (a thermal conductivity reported from a TDTR experiment, a Raman peak). The materialization of a source state carries the externally supplied value (e.g., 5.43 Å for the lattice constant of silicon, or $T = 300$ K); its discretization scheme is trivial (a singleton choice rather than a refinable mesh) but its error is still meaningful (experimental uncertainty, or definitional precision).

- *Derived states* are the outputs of operations that consume one or more upstream states. Examples: `ForceConstants[order=n]` (derived from `Potential` via `compute_force_constants[order=n]`, currently stubbed), `Frequency` and `Eigenvectors` (multi-output of `compute_dispersion`), `Linewidth`, `MeanFreeDisplacement`, the computed κ .

Both kinds follow the same machinery for materialization. The unification of source and derived states under the same “output of an operation” rule answers several otherwise awkward cases: empirical inputs (“where does the lattice constant live?”), experimental observables (“how does a measured $\kappa(T)$ enter the substrate?”), and the unmodeled abstract root in Phase 1 (“what is the silicon potential, before we differentiate it?”). All three are materializations of source states whose provenance is a single nullary operation; the second is what makes the unified theory–experiment comparison framework native rather than bolted on; the third is how Phase 1 stubs the upstream of `Potential` without yet modeling it in detail.

What “witness-as-state” means, concretely. The phrase *witness-as-state* comes from the Curry–Howard correspondence in type theory: a typed value is read as evidence (a “witness”) for a proposition. We borrow the intuition without borrowing the formal proof obligations. In our substrate a state is not the data it contains; it is the *claim* that something exists, expressed in a typed form rich enough to carry meaning.

Concretely, three layers are conflated in most computational physics codes; the substrate keeps them apart:

1. **The proposition** – “the phonon dispersion of crystalline silicon under the harmonic approximation, derived by truncating the Born–Oppenheimer Taylor expansion at second order, exists as a function $\omega(\mathbf{q}, \nu)$ from the Brillouin zone to positive reals.”
2. **The witness** – a typed object $\mathbf{s} = (\text{Dispersion}, \pi)$ that records precisely this claim. The physics type `Dispersion` encodes the kind of object being asserted to exist; the provenance π records the chain of operations (extraction, truncation, Fourier transform, eigendecomposition) that produced it. The witness carries no numerical content; it is the assertion that such an object exists with this history.
3. **The materialization** – a tuple $(s, \Sigma, x, \epsilon_{\text{disc}})$ in which x is a concrete 4-D `numpy` array of frequencies sampled on a chosen k-mesh, Σ is the discretization scheme, and ϵ_{disc} is the associated discretization error. The materialization is the numerical evidence that the witness asserts to exist.

A code that conflates these layers (treats the dispersion *as* the array) cannot distinguish “the same physics computed two different ways” from “different physics computed the same way.” A substrate that keeps them apart can: two witnesses with identical types but different provenance are different states, regardless of whether their materializations happen to have similar numerical content; one witness with two materializations on different k-meshes is the same state, regardless of how different the arrays look numerically.

This is the load-bearing distinction. It is also the answer to several otherwise hard questions. “What is the dispersion of silicon?” is the question the witness answers. “What does the dispersion of silicon look like on a $14 \times 14 \times 14$ mesh in kaldo” is the question its materialization answers. “Are these two computed dispersions the same?” resolves into two sub-questions—are the witnesses equal (a question about provenance), and are the materializations consistent within their discretization errors (a question about numerics)—each answerable without confusion with the other.

Read in this light: the abstract states described elsewhere in this document are all witnesses: they assert the existence of a particular physics object derived in a particular way. The numerical content, when present, lives in the materialization, separated from the witness by the abstract/numeric boundary. Operations between abstract states are operations between witnesses, with no numerics involved; they are essentially structural moves on a graph of typed assertions.

4.2 Symbolic operation

A symbolic operation $\mathcal{O}$ is a typed map between abstract states,

$$\mathcal{O} : \tau_{\text{in}_1} \times \tau_{\text{in}_2} \times \cdots \rightarrow (\tau_{\text{out}_1}, \dots, \tau_{\text{out}_m}), \quad (2)$$

equipped with three pieces of metadata:

- a *symbolic formula* $\mathcal{F}_{\mathcal{O}}$ stating what the operation computes, encoded as a sympy expression for closed-form ops or a sympy.Eq for implicit ones (eigenvalue equations, linear systems). Source operations (`provide_*`) carry $\mathcal{F}_{\mathcal{O}} = \emptyset$;
- an *approximation error formula* $\epsilon_{\mathcal{O}}$, a symbolic expression in physical small parameters (e.g., λ for the strength of a perturbation, T/T^* for a Sommerfeld expansion). Exact operations have $\epsilon_{\mathcal{O}} = 0$; approximating operations have $\epsilon_{\mathcal{O}} = O(p^k)$ for some parameter p and order k ;
- a possibly empty set of *algorithmic conventions* (per §4.2) and *parameters* (dimensioned but unit-free at the abstract layer).

Multi-output operations (e.g. `compute_dispersion` producing `Frequency` and `Eigenvectors` together) are first-class. The formula $\mathcal{F}_{\mathcal{O}}$ is the substrate’s machine-readable claim about what the operation produces: adapter conformance can be expressed as “the adapter’s runtime behavior matches $\mathcal{F}_{\mathcal{O}}$ under its declared unit and convention choices,” in principle automatable rather than reverse-engineered from kernels.

Operation identity is parameterized. Several operations in the substrate carry method choices that affect physics: the scattering processes included (3-phonon only, or 3-phonon plus isotopic, or with boundary scattering), the Boltzmann solver (relaxation-time approximation, iterative, direct inversion), the broadening scheme (Gaussian, Lorentzian, adaptive), and so on. The substrate treats these as part of the operation’s *identity*, not as runtime configuration that gets discarded.

Concretely, an operation is identified by the tuple (operation name, parameters affecting physics). Two invocations of `compute_scattering_rates` with different `scattering_processes` are different operations; they appear differently in the provenance, and they produce abstract states that are formally distinct (different type metadata, different total approximation error). The *name* `compute_scattering_rates` is the same; the *parameterized identity* differs.

This convention has three consequences. First, the vocabulary of operations remains small (one entry per named transformation, not one entry per parameter combination), but the provenance remains expressive (because provenance records the full parameterized identity). Second, output states’ types are mildly dependent: `ScatteringRates` is parameterized by the scattering-processes choice, even though the substrate does not require Python’s type system to enforce this dependence. Third, the substrate is structurally compatible with dependent type theory: when transliterated to a system like Lean, the parameter values lift to type-level indices and the dependence becomes formal.

4.3 Abstract workflow

An abstract workflow is a finite directed acyclic graph $W = (V, E)$ where V is a set of abstract states and E is a set of symbolic operations. The graph is rooted at one or more *source* states (typically empirical inputs such as the crystal structure) and terminates at one or more *observable* states (e.g., the lattice thermal conductivity).

4.4 Discretization scheme

A discretization scheme Σ is a finite tuple of parameters $(N_1, \dots; h_1, \dots)$ specifying how a continuum quantity is sampled. Examples include the k-point mesh density, the supercell size, the finite-difference displacement step, the broadening width, and the integration order. The scheme also carries an error model $\epsilon_\Sigma(N_1, \dots; h_1, \dots)$ giving the asymptotic dependence of the discretization error on these parameters.

4.5 Materialization

A materialization m of an abstract state $s = (\tau, \pi)$ is a tuple $(s, \Sigma, x, U, \epsilon_{\text{disc}})$ where:

- s is the abstract state being materialized (a back-reference);
- Σ is a discretization scheme;
- x is a concrete numerical object whose representation is compatible with τ (e.g., a 4-D array if $\tau = \text{Dispersion}$, a scalar if $\tau = \text{ThermalConductivity}$);
- U is the *unit declaration*: a mapping from each dimensionful quantity in τ to the unit in which it is expressed in x (e.g., for `Dispersion`, `{frequency: linear THz, group_velocity: Å·THz}`). The adapter that produced m supplies U ;
- $\epsilon_{\text{disc}} = \epsilon_\Sigma$ instantiated at the chosen parameters.

The materialization carries no approximation error of its own; that information is recovered via s and its provenance π .

Why U is part of the materialization. Different codes express the same physics in different units (e.g., `kaldo`’s linewidth is in angular-frequency THz while `phono3py`’s is in linear-frequency THz, a factor of 4π ; `kaldo`’s heat capacity is in J/K, `phono3py`’s in eV/K, a factor of e). A materialization without U is unsafe to compare against another; with U , the substrate can convert both to a canonical unit before any comparison and surface unit mismatches as adapter conformance failures rather than as numerical mysteries.

4.6 Observable extraction and comparison protocols

Cross-adapter agreement is checked at the observable layer, not at the raw materialization layer. An *observable* is a pure function $Q : x \mapsto y$ that extracts a physically meaningful quantity from the numerical content x of a materialization (a contraction, a peak position, an integrated quantity). Each abstract state declares its observables; each observable carries a *comparison protocol* specifying how two extractions y_1, y_2 from materializations of the same abstract state by different adapters are compared—typically a tolerance, but possibly a richer relation (e.g., “equal up to a permutation of degenerate modes” or “not directly comparable, contraction required”).

The protocol records the substrate’s claim about which extractions are stable across adapter choices and which are not. A contracted scalar like $\kappa^{\alpha\beta}$ or $\Sigma\Gamma_{q\nu}$ has a tight tolerance: disagreement is a real discrepancy that must be diagnosed. A per-mode array whose values depend on

the summation order in the underlying sum (e.g., per-mode $\Gamma_{q\nu}$ in a Brillouin-zone integration that one adapter handles by symmetry-reduced triplets and another handles by enumeration) is explicitly marked as not-directly-comparable: numerical disagreement at the mode level is expected and not a substrate failure, even though the contracted total agrees.

Why this is the right level. Two adapters that disagree at the observable layer disagree about physics. Two adapters that disagree at the raw-materialization layer but agree at the observable layer disagree only about bookkeeping—the order in which a sum is decomposed, the basis in which a tensor is written. The substrate’s job is to make this distinction structural: by declaring which observables are adapter-invariant, the substrate states what it is asking for and what it is not.

4.7 Materialization functor

A materialization functor $\mathcal{F}$ is associated with a particular code (kaldo, phono3py, ShengBTE). Given an abstract workflow W and a discretization scheme Σ ,

$$\mathcal{F}(W, \Sigma) \tag{3}$$

produces a directed graph of materializations whose vertices are in one-to-one correspondence with a subset $V_{\mathcal{F}} \subseteq V(W)$ and whose edges are the computational realizations of the symbolic operations restricted to $V_{\mathcal{F}}$. Different codes implement different functors over the same abstract workflow; their materialized graphs are aligned by their shared abstract template.

4.8 Total error

The total error on the materialization m of an abstract state s is

$$\epsilon_{\text{total}}(m) = \text{compose}(\epsilon_{\text{approx}}(s), \epsilon_{\text{disc}}(m)). \tag{4}$$

The composition rule depends on the operation type but is in all cases derivable from the substrate’s typed metadata. The two contributions $\epsilon_{\text{approx}}(s)$ and $\epsilon_{\text{disc}}(m)$ are recovered separately, so a result that is converged in discretization but uncertain in approximation is distinguishable from one that is fully approximated but undersampled.

5 Lean-compatibility disciplines

“Lean-compatible” is structural rather than syntactic. The substrate’s Python implementation will not be Lean code, but it should be *shaped* so that the Lean version is a transliteration rather than a redesign. We commit to three disciplines that protect this option.

1. **Closed unions for physics types.** The registry of physics types (`Potential`, `ForceConstants`, `Dispersion`, `ScatteringRates`, ...) is exhaustive and closed—implemented in Python via sealed enums or tagged unions, not via open class inheritance. Lean represents these as inductive types with a fixed list of constructors; closed unions in Python preserve this structure exactly. Open inheritance would require redesign at the Lean stage.
2. **No physics invariants encoded in Python types.** Properties such as “a force-constant tensor is symmetric under the appropriate index exchanges” or “the eigenvectors of a dynamical matrix are orthonormal” are documented as informal invariants on each type but *not* enforced by the Python type system. In Lean these invariants become formal proof obligations attached to the corresponding types. Trying to encode them in Python (via runtime assertions or pydantic validators) creates a layer of checks that does not transliterate and risks coupling the implementation to Python-specific idioms.

3. **Operation parameters always recorded in output state metadata.** When an operation carries parameters that affect physics (§4.2), Python’s type system will not enforce the dependence of the output type on those parameters, but the output state’s metadata must still record them. In Lean these parameters lift to type-level indices, and the type-level dependence is enforced formally; in Python they are runtime metadata that downstream operations and provenance machinery treat as part of the operation’s identity.

These disciplines are inexpensive to follow from day one and prohibitive to retrofit. They keep the transliteration cost low without committing the substrate to Lean tooling we do not yet need.

6 First Demonstration: Lattice Thermal Transport

We instantiate the substrate on the canonical workflow for lattice thermal conductivity κ , computed from second- and third-order interatomic force constants via the linearized Boltzmann transport equation.

6.1 The abstract DAG

The abstract DAG has 12 nodes (states / observables) and 11 edges (operations). Two nodes (**Potential**, **Temperature**) are sources, produced by nullary `provide_*` operations.

Nodes.

- **Potential**: the abstract root, materialized in Phase 1 as a stubbed source state (the silicon potential, the LAMMPS-Tersoff potential, or the QE-PBE potential, as an opaque label);
- **Temperature**: source state, scalar T ;
- **ForceConstants[order=2]**, **ForceConstants[order=3]**: harmonic and cubic interatomic force constants $\Phi^{(n)}$;
- **DynamicalMatrix**: the Bloch sum $D_{ij}(\mathbf{q})$;
- **Frequency**: $\omega(\mathbf{q}, \nu)$, indices $(\mathbf{q}, \nu)$;
- **Eigenvectors**: $e(\mathbf{q}, \nu)$, indices $(i, \mathbf{q}, \nu)$;
- **GroupVelocity**: $v^\alpha(\mathbf{q}, \nu)$, indices $(\alpha, \mathbf{q}, \nu)$;
- **HeatCapacity**: $c(\mathbf{q}, \nu, T)$, indices $(\mathbf{q}, \nu)$;
- **Linewidth**: $\Gamma(\mathbf{q}, \nu, T)$, indices $(\mathbf{q}, \nu)$;
- **MeanFreeDisplacement**: $F^\alpha(\mathbf{q}, \nu, T)$, indices $(\alpha, \mathbf{q}, \nu)$;
- **ThermalConductivity**: $\kappa^{\alpha\beta}(T)$, indices (α, β) .

Edges. Verb-headed names; each carries a sympy formula:

- `provide_potential`: $() \rightarrow \text{Potential}$;
- `provide_temperature`: $() \rightarrow \text{Temperature}$;
- `compute_force_constants[order=2]`, `[order=3]`: $\text{Potential} \rightarrow \text{ForceConstants[order}=n]$ (stubbed in Phase 1);
- `compute_dynamical_matrix`: $\text{ForceConstants[order=2]} \rightarrow \text{DynamicalMatrix}$;

states are exposed via Python attributes on the `Phonons` object.

- **phono3py** (C/Python). Computes lattice thermal conductivity from third-order force constants, with extensive support for finite-displacement and DFPT pipelines. Intermediate quantities are exposed in HDF5 outputs.
- **ShengBTE** (Fortran/MPI). Solves the full linearized BTE with iterative self-consistency, exposing intermediate quantities in plain-text output files.

For each code, a thin adapter reads the code’s natural output (Python attribute, HDF5 dataset, or text file), constructs the corresponding materialization, and tags it with the abstract state it realizes. The first concrete demonstrable result of the substrate is to run all three codes on a single material (silicon) at a single discretization, and to display the three materialized DAGs aligned to the shared abstract template.

6.3 Phase 1 scientific capabilities

Phase 1 yields two immediate scientific capabilities, each of which is an artifact the field does not currently produce. A third capability is the long-term goal toward which the substrate is oriented but is deferred to Phase 2.

- **Cross-code reconciliation at canonical intermediate states.** When `kaldo` and `ShengBTE` disagree on a final κ , the substrate localizes the divergence to the first abstract state at which the materializations stop agreeing, and attributes it to a specific operation along the provenance. This is a structural diagnosis: the substrate identifies where two codes do something different, regardless of the magnitude of the difference.
- **A unified framework for theory-experiment comparison.** The same abstract DAG accepts experimental measurements as additional materializations of parameter states, with the same alignment machinery as for cross-code comparison. Phase 1 demonstrates the framework on at least one experimental dataset (e.g., a measured $\kappa(T)$ curve for silicon).

Deferred to Phase 2:

- **Decomposed error bounds on κ .** The substrate is designed to eventually produce the central value of κ together with a symbolic decomposition of its total error into the truncation error of the perturbation series and the discretization error of the chosen mesh. The architecture supports this; the implementation of the composition algebra is a research problem reserved for Phase 2.

7 Implementation layout

The Python package `omai` factors the substrate into three top-level sub-packages plus one per domain:

- `omai.abstract`: substrate machinery for the abstract world. `State`, `Operation`, `Observable`, `Parameter`, `Dimension` (class), `PhysicsType` (closed enum), and `topological_order`. Code-agnostic and domain-agnostic.
- `omai.materialization`: the bridge. `Unit` class with the unit registry plus strict same-dimension `conversion_factor`; `StateAdapterSpec` and `OperationAdapterSpec` with cross-state / cross-operation comparison helpers. Code-agnostic.
- `omai.thermal_transport`: the lattice-thermal-transport domain.

- `symbolic.py`: the 12 nodes and 11 edges of §6 as `State` and `Operation` instances, with sympy formulas on every derived edge.
- `materialized/kaldo.py`, `materialized/phono3py.py`: per-code adapter specs (one file per code), each declaring units, conventions, and discretization choices for the abstract DAG.

The split mirrors the substrate’s architectural commitment: `abstract` and `materialization` are the substrate’s two worlds plus its bridge; `thermal_transport` is the first domain instantiated against them. Future domains (electronic transport, optical response, etc.) follow the same two-file pattern in their own folder; no change to `abstract` or `materialization` is required to add one.

8 Open Questions and Deferred Decisions

We list the architectural questions that remain open and the principled grounds on which they were deferred.

8.1 Algebra of error composition (deferred to Phase 2)

The substrate is designed to carry error formulas as symbolic expressions, but Phase 1 does not implement the algebra by which they compose. Composing two errors in independent small parameters ($O(\lambda^4) + O(N^{-2})$) is well-defined; composing two errors in the same parameter requires care; composing constant-offset errors (e.g., basis-set incompleteness) with asymptotic ones is yet another case. The right algebra is itself a research subproblem and will be developed alongside concrete worked examples in Phase 2. The Phase 1 type system reserves the slots required for error formulas (operation metadata, discretization error model, materialization total error) but populates them with placeholders rather than computed compositions.

8.2 When to lift to Lean

We have committed to keeping the substrate Lean-compatible without being Lean-dependent. The question of when (and whether) to actually project the abstract layer into Lean is deferred until the Python prototype has stabilized. A reasonable trigger for this decision is the moment at which we have enough concrete examples that designing Lean types in isolation becomes counterproductive.

8.3 Provenance equivalence

We have committed to provenance as essential to identity. This means two states with the same type but different histories are formally distinct. In some cases, however, the two histories are *equivalent*: they produce identical physics under a theorem we can prove (e.g., Fourier transform of a real symmetric matrix and direct frequency-domain assembly should yield the same `Dispersion`). The substrate must provide a mechanism for declaring such equivalences, either as explicit axioms or as proven theorems. The mechanism’s design is deferred to Day 2 because we want concrete examples before designing the equivalence calculus.

8.4 Sprint 1 scope

The first concrete deliverable is a Python implementation. We have considered three scopes:

1. *Narrowest*: kaldo only, harmonic sub-DAG (force constants $\rightarrow$ dynamical matrix $\rightarrow$ dispersion). 4–5 states, fastest to ship.

2. *Recommended*: kaldo only, full thermal-conductivity DAG. 10–15 states, end-to-end on silicon. The substrate skeleton plus one full adapter.
3. *Broader*: kaldo plus phono3py, full DAG. Validates the cross-code story earlier but adds adapter work in parallel with substrate design.

The recommended scope (option 2) is the smallest scope that is still a credible proof of concept of the substrate, and the choice of one adapter (kaldo) lets us validate the substrate design before generalizing. Sprint 2 adds phono3py and ShengBTE adapters, at which point the cross-code demo of §6 becomes real.

8.5 Implementation language

We have committed to Python for Sprint 1. The trade-offs were:

- **Python (chosen)**. All target codes have Python bindings or wrappable interfaces. The Python type system, augmented with `pydantic` or `attrs` and runtime checks, is sufficient for the substrate’s first-cut type discipline. The development speed is dominant, since adapter work is the bulk of the time cost.
- *Typed compiled language (Rust, OCaml, TypeScript)*. Stronger type guarantees from the start. Rejected for Sprint 1 because the foreign-function interface to existing Python codes adds friction proportional to the substrate’s surface, and we want substrate iteration to be cheap.
- *Lean*. Considered as a future projection target and deferred. Not the right tool for running scientific computations; reserved as a verification target.

9 Outlook

The substrate is an architectural commitment, not an implementation. The implementation that follows is scoped to a Python skeleton plus adapters for three thermal-transport codes. The artifact that motivates the project—a paper-worthy demonstration of cross-code reconciliation and decomposed error bounds for lattice thermal conductivity—is downstream of the substrate but is the substrate’s first proof-of-value.

Three longer-term directions follow naturally from the design:

- **Verification**. Project the abstract layer into Lean (via PhysLean or alongside it), allowing approximation theorems to be formally proved rather than asserted. The provenance mechanism makes this projection well-defined: each operation in π corresponds to a Lean lemma whose statement is the operation’s error formula.
- **Theory-experiment integration**. Extend the materialization functor to ingest experimental observables. The substrate’s graph-alignment machinery applies uniformly, so a measured κ becomes another materialization of the abstract κ , and “does the theory match experiment” becomes graph-alignment with one functor being empirical.
- **Grounded LLM agents**. An LLM operating over typed abstract states (rather than text tokens) constructs workflows by emitting symbolic operations. The substrate type-checks each emission, eliminating the drift problem that plagues current text-token agents. The action space is finite, typed, and physically meaningful.

The unifying claim of the design is that scientific computing has been organized around the wrong atomic unit. The instruction-in-an-input-file unit is too low; the natural-language description is too high; the equation-in-a-paper unit is too disconnected from execution. The right unit, at least for materials science, is the symbolic operation between typed physics states. Once that commitment is made, much of the rest of the architecture writes itself.

This document records the architectural state of the openmaterials-ai project as of May 10, 2026. It is intended as the working reference for the project; subsequent revisions should update the relevant sections rather than appending changelogs.